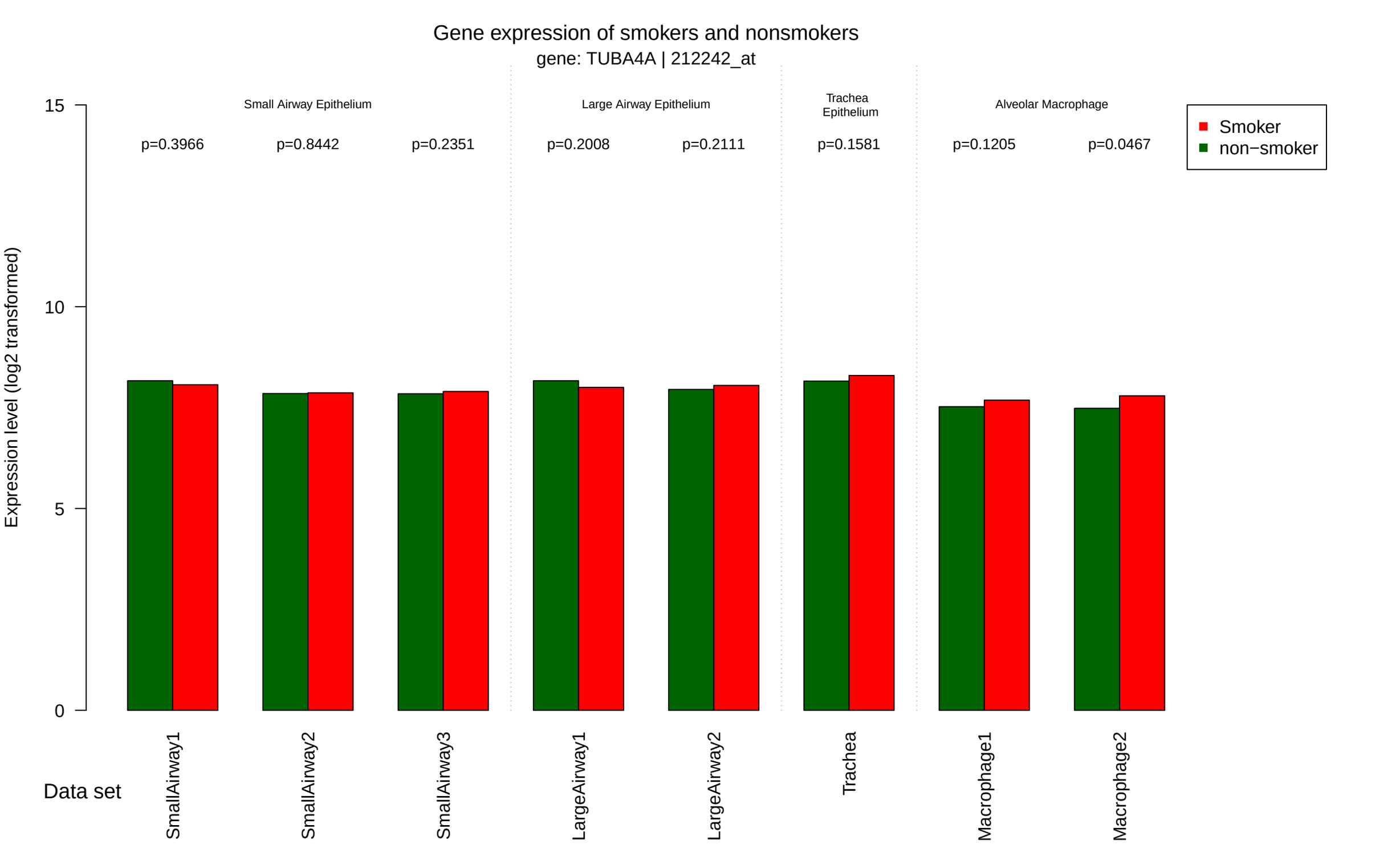
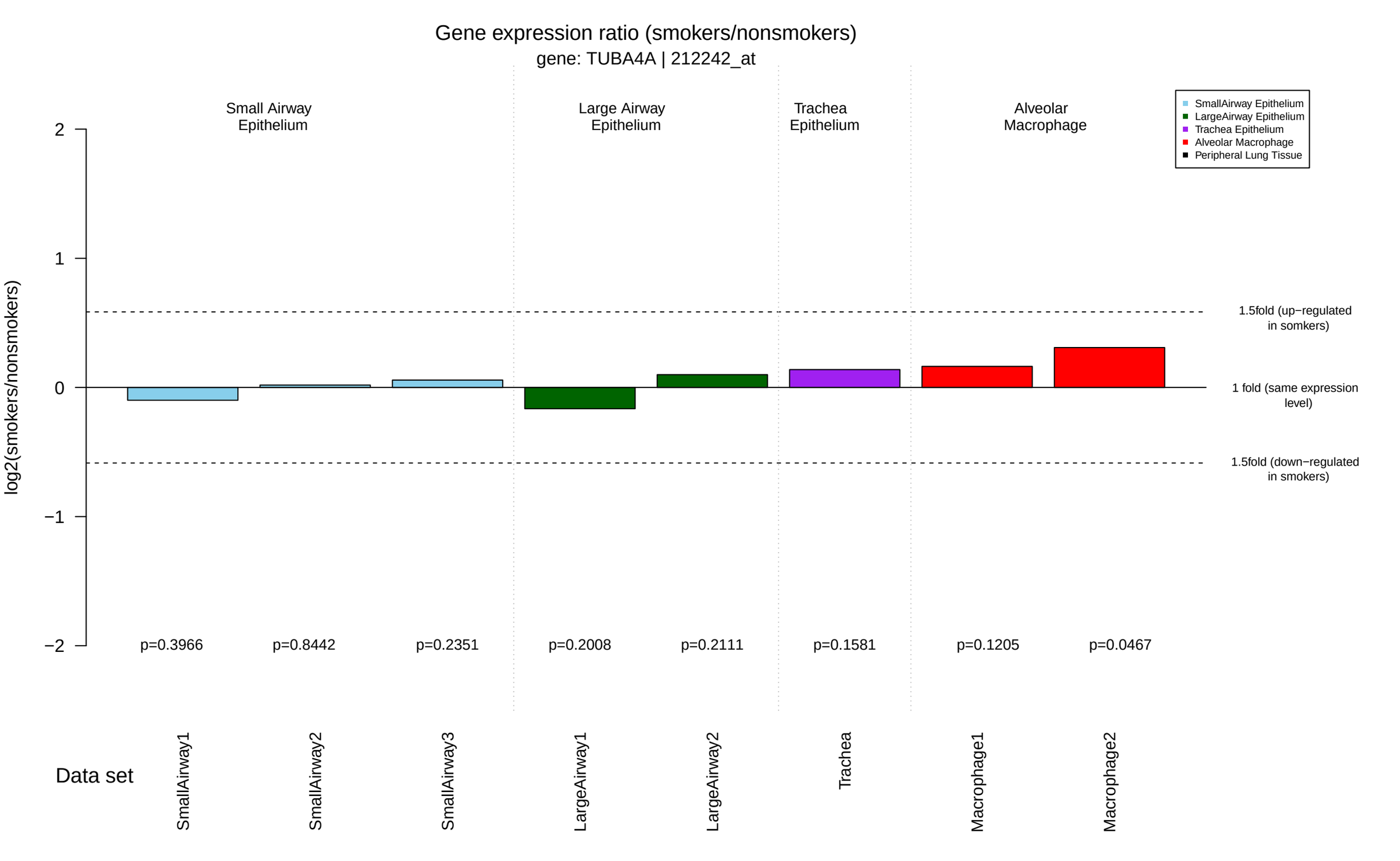
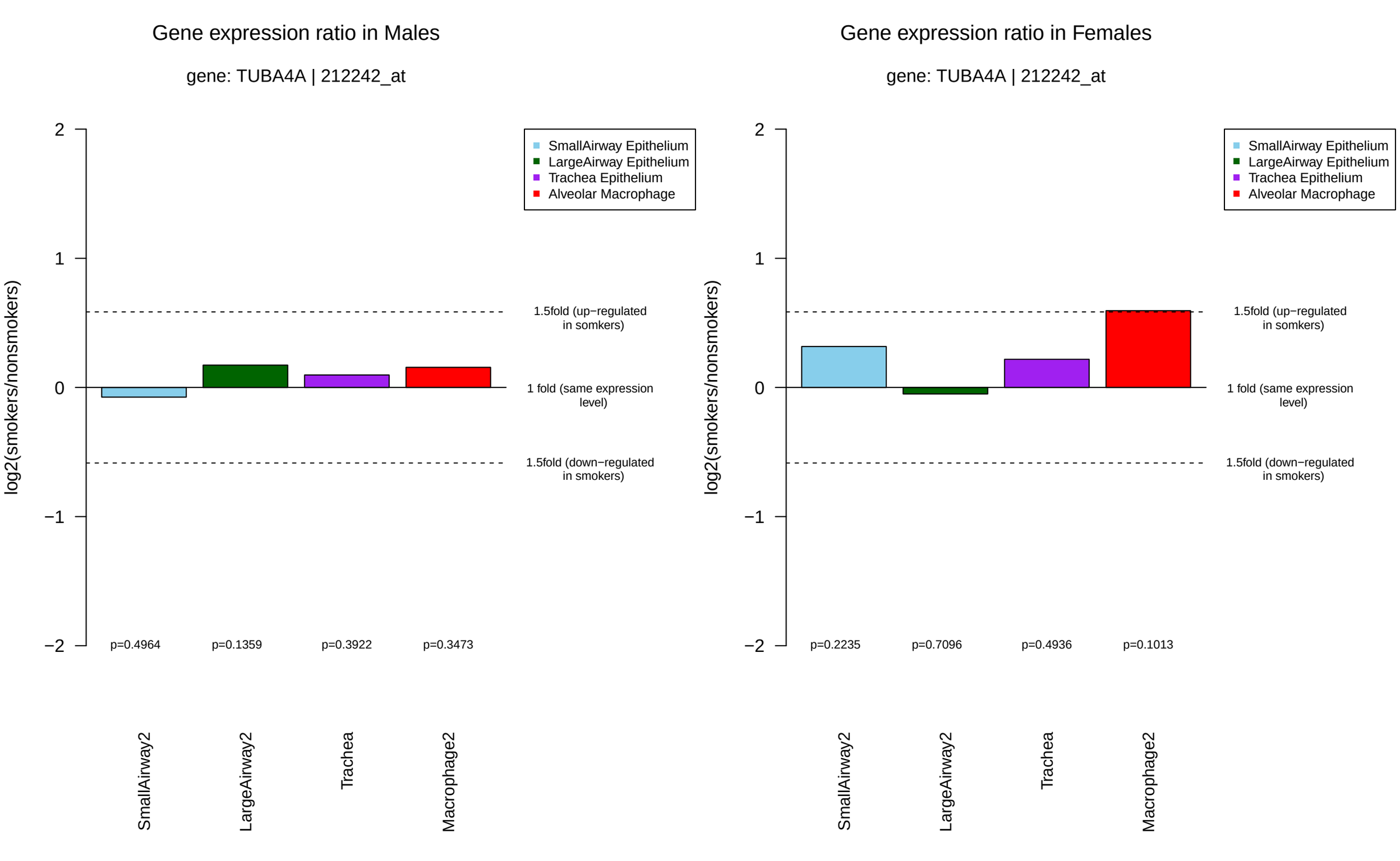
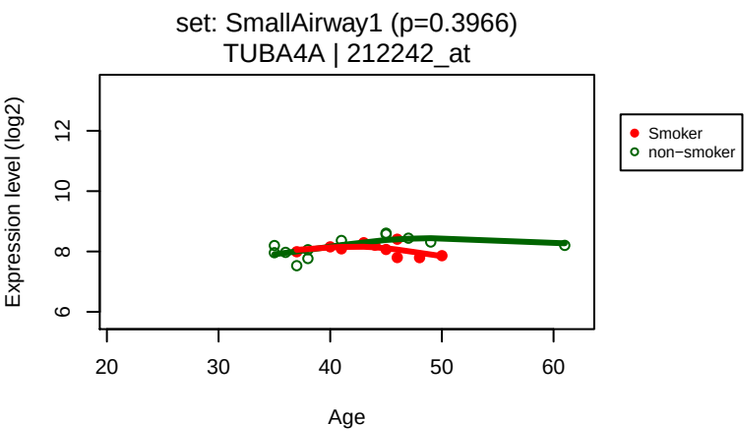
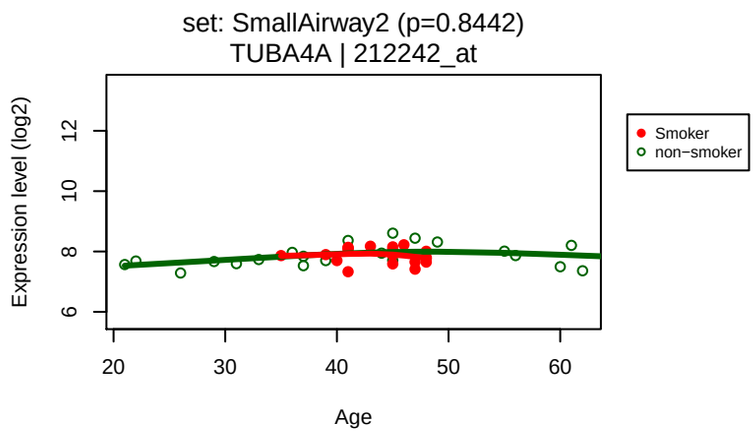
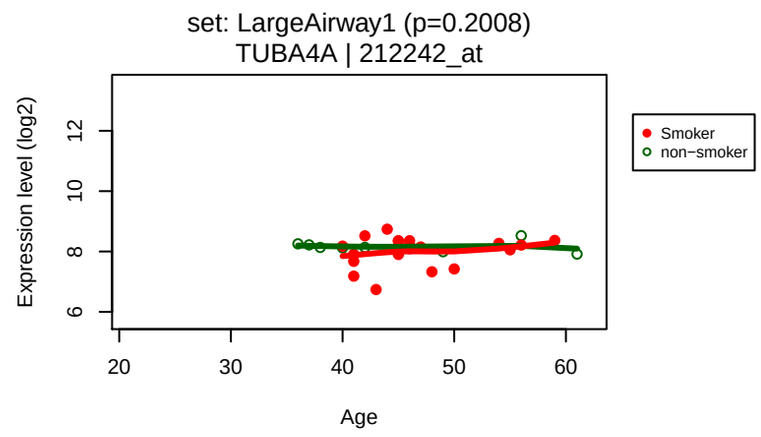

Gene expression of smokers and nonsmokers
gene: TUBA4A | 212242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3966
p=0.8442
p=0.2351
p=0.2008
p=0.2111
p=0.1581
p=0.1205
p=0.0467
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

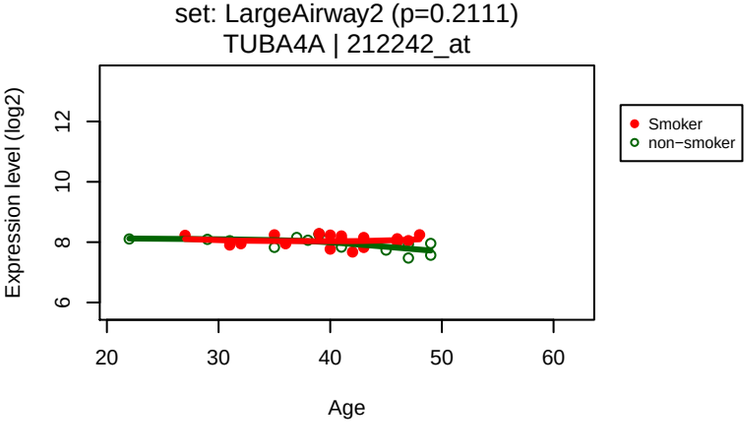

Gene expression ratio (smokers/nonsmokers)
gene: TUBA4A | 212242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3966
p=0.8442
p=0.2351
p=0.2008
p=0.2111
p=0.1581
p=0.1205
p=0.0467
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

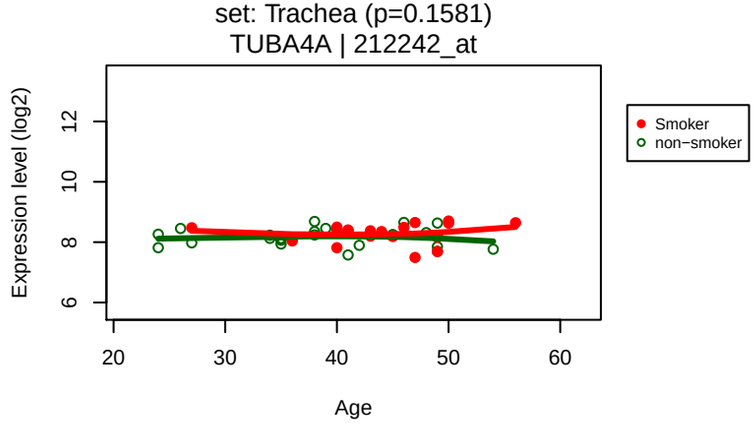

Gene expression ratio in Males
gene: TUBA4A | 212242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4964
p=0.1359
p=0.3922
p=0.3473
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TUBA4A | 212242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2235
p=0.7096
p=0.4936
p=0.1013
SmallAirway2
LargeAirway2
Trachea
Macrophage2

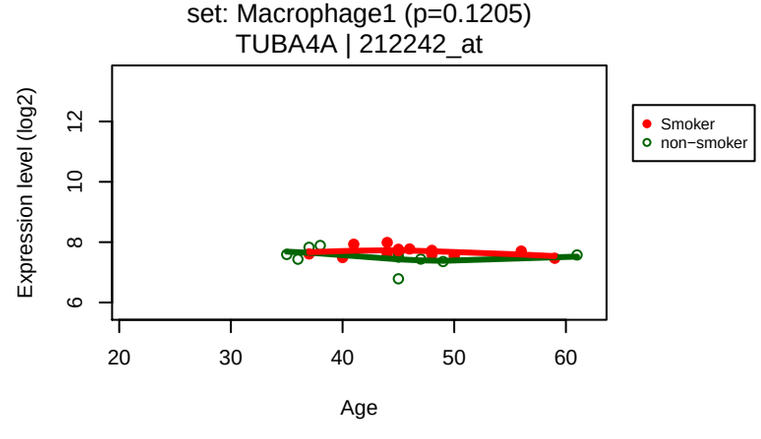

set: SmallAirway1 (p=0.3966)
TUBA4A | 212242_at
Expression level (log2)
Age
Smoker
non−smoker

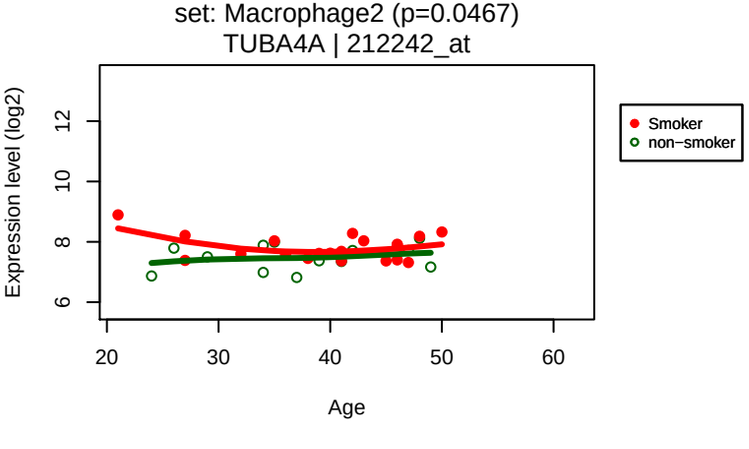

set: SmallAirway2 (p=0.8442)
TUBA4A | 212242_at
Expression level (log2)
Age
Smoker
non−smoker

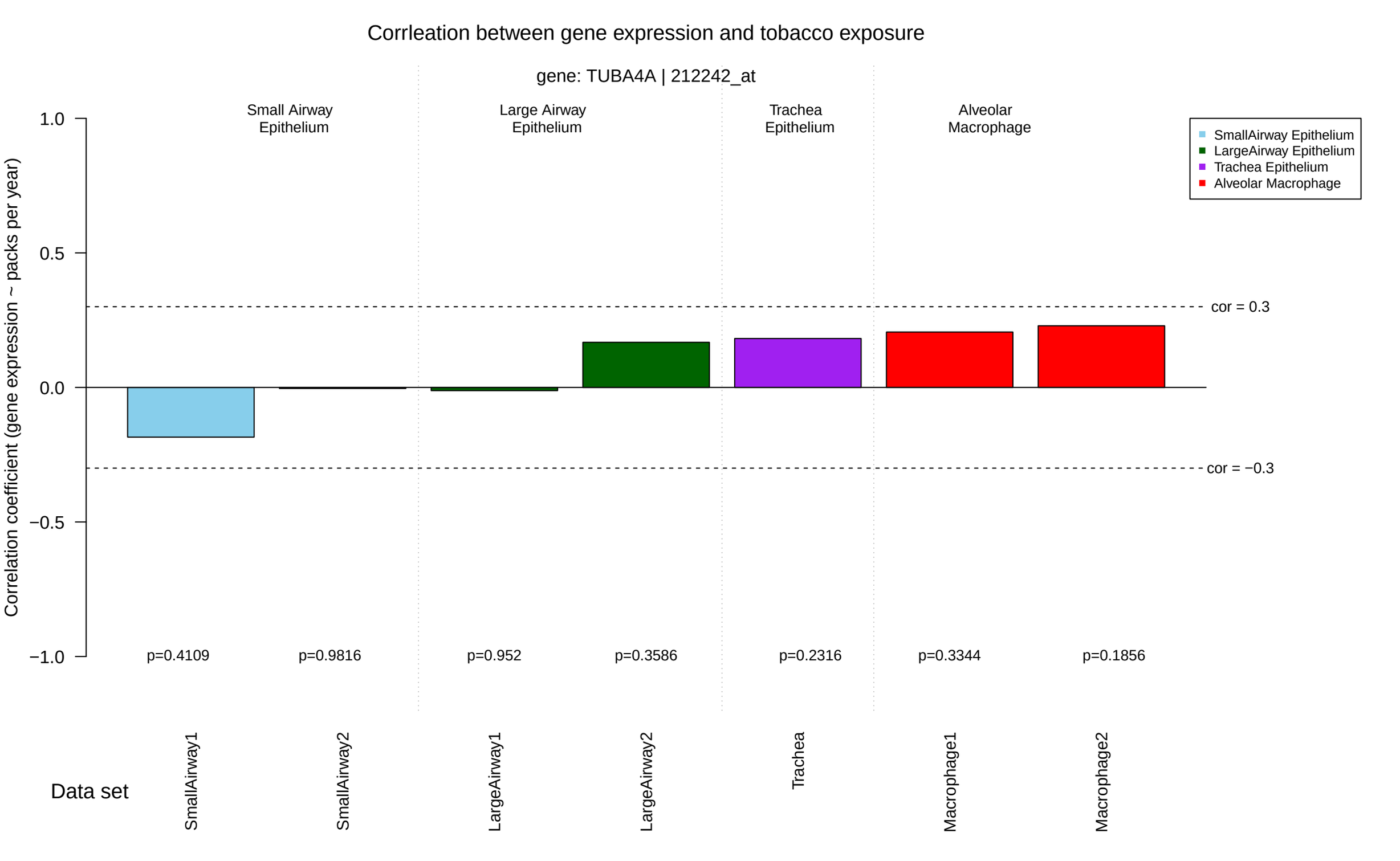

set: LargeAirway1 (p=0.2008)
TUBA4A | 212242_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2111)
TUBA4A | 212242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1581)
TUBA4A | 212242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1205)
TUBA4A | 212242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0467)
TUBA4A | 212242_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TUBA4A | 212242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4109
p=0.9816
p=0.952
p=0.3586
p=0.2316
p=0.3344
p=0.1856
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2